CENTRO UNIVERSITÁRIO BELAS ARTES DE SÃO PAULO

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Relatório Técnico: Desenvolvimento de API com Framework FastAPI

SÃO PAULO

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1. Introdução

Nesta atividade, foi desenvolvida uma API utilizando o framework **FastAPI** em linguagem Python 3.12. O objetivo foi configurar um ambiente de desenvolvimento isolado, instalar as dependências necessárias e criar rotas que retornem dados em formato JSON.

3. Desenvolvimento e Configuração

3.1. Ambiente Virtual (venv)

Foi criado um ambiente virtual para isolar as bibliotecas do projeto.

- **Comando utilizado:** `python -m venv myenv`

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    powershell

PS C:\Users\Manuela Lobo\Downloads\projeto-fastapi> & "c:\Users\Manuela Lobo\Downloads\projeto-fastapi\myenv\Scripts\Activate.ps1"
(myenv) PS C:\Users\Manuela Lobo\Downloads\projeto-fastapi>
```

3.2. Instalação das Dependências

Com o ambiente ativo, foram instalados o **FastAPI** (framework) e o **Uvicorn** (servidor ASGI).

- **Comando utilizado:** `pip install fastapi uvicorn`

```
(myenv) PS C:\Users\Manuela Lobo\Downloads\projeto-fastapi> pip install fastapi uvicorn
Collecting fastapi
  Using cached fastapi-0.135.1-py3-none-any.whl.metadata (30 kB)
Collecting uvicorn
  Using cached uvicorn-0.42.0-py3-none-any.whl.metadata (6.7 kB)
Collecting starlette>=0.46.0 (from fastapi)
  Using cached starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
Collecting pydantic>=2.7.0 (from fastapi)
  Using cached pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting typing-extensions>=4.8.0 (from fastapi)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi)
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi)
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting click>=7.0 (from uvicorn)
  Using cached click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting h11>=0.8 (from uvicorn)
  Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting colorama (from click>=7.0->uvicorn)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting annotated-types>=0.6.0 (from pydantic>=2.7.0->fastapi)
  Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.41.5 (from pydantic>=2.7.0->fastapi)
  Using cached pydantic_core-2.41.5-cp312-cp312-win_amd64.whl.metadata (7.4 kB)
Collecting anyio<5,>=3.6.2 (from starlette>=0.46.0->fastapi)
  Using cached anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
Collecting idna>=2.8 (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi)
  Using cached idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Using cached fastapi-0.135.1-py3-none-any.whl (116 kB)
Using cached uvicorn-0.42.0-py3-none-any.whl (68 kB)
Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Using cached click-8.3.1-py3-none-any.whl (108 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached pydantic-2.12.5-py3-none-any.whl (463 kB)
Using cached pydantic_core-2.41.5-cp312-cp312-win_amd64.whl (2.0 MB)
Using cached starlette-0.52.1-py3-none-any.whl (74 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Using cached typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Using cached annotated_types-0.7.0-py3-none-any.whl (13 kB)
Using cached anyio-4.12.1-py3-none-any.whl (113 kB)
```

```
Collecting pydantic>=2.7.0 (from fastapi)
  Using cached pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting typing-extensions>=4.8.0 (from fastapi)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi)
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi)
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting click>=7.0 (from uvicorn)
  Using cached click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting h11>=0.8 (from uvicorn)
  Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
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  Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.41.5 (from pydantic>=2.7.0->fastapi)
  Using cached pydantic_core-2.41.5-cp312-cp312-win_amd64.whl.metadata (7.4 kB)
Collecting anyio<5,>=3.6.2 (from starlette>=0.46.0->fastapi)
  Using cached anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
Collecting idna>=2.8 (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi)
  Using cached idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Using cached fastapi-0.135.1-py3-none-any.whl (116 kB)
Using cached uvicorn-0.42.0-py3-none-any.whl (68 kB)
Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Using cached click-8.3.1-py3-none-any.whl (108 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached pydantic-2.12.5-py3-none-any.whl (463 kB)
Using cached pydantic_core-2.41.5-cp312-cp312-win_amd64.whl (2.0 MB)
Using cached starlette-0.52.1-py3-none-any.whl (74 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Using cached typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Using cached annotated_types-0.7.0-py3-none-any.whl (13 kB)
Using cached anyio-4.12.1-py3-none-any.whl (113 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Using cached idna-3.11-py3-none-any.whl (71 kB)
Installing collected packages: typing-extensions, idna, h11, colorama, annotated-types, annotated-doc, typing-inspection, pydantic-core, click, anyio, uvicorn, starlette, pydanti
c, fastapi
Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.12.1 click-8.3.1 colorama-0.4.6 fastapi-0.135.1 h11-0.16.0 idna-3.11 pydantic-2.12.5 pydantic-core-2.41.5
inspection-0.4.2 uvicorn-0.42.0
```

3.3. Código da Aplicação (`main.py`)

O código foi estruturado com duas rotas do tipo GET. Uma para a raiz do servidor e outra para identificação da aluna.

```python
main.py
projeto-fastapi > main.py > ler_nome
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
def read_root():
    return {"Hello": "World"}


@app.get("/identificacao")
def ler_nome():
    return {
        "aluno": "Manuela Lobo",
        "rm": "24109798",
        "projeto": "Desafio FastAPI - ADS"
    }
```

4. Testes e Resultados

4.1. Execução do Servidor

O servidor foi iniciado utilizando o comando `uvicorn main:app --reload`.

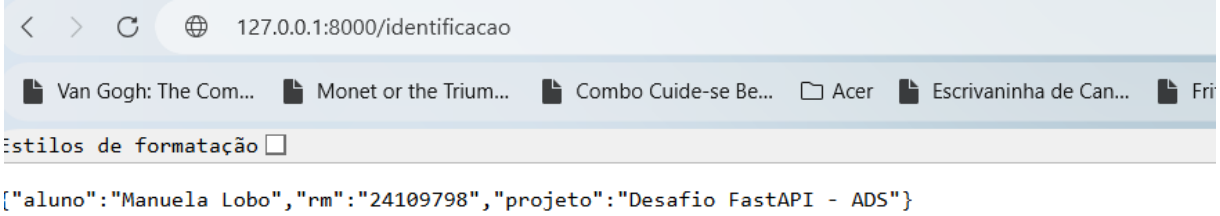
- **Print 4:**

```
(myenv) PS C:\Users\Manuela Lobo\Downloads\projeto-fastapi> uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\Manuela Lobo\\Downloads\\projeto-fastapi']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [32852] using StatReload
INFO:     Started server process [25148]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

4.2. Retorno das Rotas (Navegador/Insomnia)

Abaixo, os resultados obtidos ao acessar os endereços da API:

- **Rota Raiz (`/`):** Exibe `{"Hello": "World"}`.
- **Rota Identificação (`/identificacao`):** Exibe o nome completo e RM.




```
{"aluno":"Manuela Lobo","rm":"24109798","projeto":"Desafio FastAPI - ADS"}
```


5. Conclusão

A atividade permitiu compreender o fluxo de criação de uma API moderna, desde a separação do ambiente de desenvolvimento até a entrega de dados via protocolo HTTP. Todos os requisitos do enunciado foram atendidos com sucesso.

Link do vídeo: https://youtu.be/-zvxmKnxBml